THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following represents 3 molecules of oxygen? 1

(a) O_6 **(b)** $6O$ **(c)** $3\,O$ **(d)** $3O_2$

2. Brittle yellow solid that burns with blue flame to give a choking gas is: 1

(a) iron **(b)** phosphorus **(c)** sulphur **(d)** carbon

3. Assertion (A): Air is a mixture and not a compound. Reason (R): The proportion of gases in air can vary. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A student heated copper powder and the colour changed to black. Which statement is correct? 1

(a) copper became a mixture **(b)** copper turned into carbon
(c) a compound copper oxide formed **(d)** an alloy was formed

5. Which of these properties best identifies a metalloid? 1

(a) shows properties of both metals and non-metals
(b) is a liquid
(c) is magnetic
(d) is a gas

6. Assertion (A): The properties of sodium chloride are very different from those of sodium and chlorine. Reason (R): A compound has properties different from its constituent elements. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A substance X conducts electricity in the solid state and is malleable. X is most likely: 1

(a) a compound of non-metals **(b)** a non-metal
(c) a metal **(d)** an inert gas

8. The use of silicon in computer chips depends on it being a: 1

(a) good insulator **(b)** radioactive element
(c) gas **(d)** semiconductor

9. Assertion (A): A heterogeneous mixture can be separated by physical methods. Reason (R): The components of a mixture lose their individual properties. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. A metal oxide is dissolved in water and turns red litmus blue. The metal oxide is: 1

(a) acidic **(b)** inert **(c)** neutral **(d)** basic in nature

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are cooking utensils made of stainless steel preferred over iron? 2

12. Differentiate between a pure substance and a mixture using milk and distilled water. 2

13. Why does a mixture of sulphur and iron filings show yellow and grey colours, but the compound does not? 2

14. Why does a piece of sodium float and react vigorously with water? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the number of atoms in a molecule determines the formula with H2O2 and H2O as examples. 3

16. State three observations when magnesium ribbon burns in air and name the product and its type. 3

17. Why do metals like potassium and sodium not occur free in nature? 3

OR

An alloy contains 90 per cent copper and 10 per cent tin. Find the mass of copper in 500 g of the alloy and the mass of tin. 3

18. Why does rusting of iron produce a new substance but dissolving sugar in water does not? 3

OR

Differentiate between combustion of magnesium and of sulphur in terms of the nature of the oxides. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the properties of metals, non-metals and metalloids, giving three exceptions and uses of each category. 5

OR

Explain how water treatment plants make use of physical separation techniques: sedimentation, filtration, chlorination and distillation. Classify each as physical or chemical. 5

20. A mixture contains sand, salt, iron filings and water. Draw up a flow chart in words to separate all components, including the property used at each step. 5

OR

Analyse the statement 'All compounds are pure substances but all pure substances are not compounds' with examples. 5

END OF QUESTION PAPER — Total: 40 Marks